

**MAT 426, Class 16 Homework: QR Factorization via Householder**  
(Due Thursday, April 9, 2026)

1. For each of the following matrices  $A$  find the QR factorization of  $A$  using Householder matrices.

(a)  $A = \begin{bmatrix} 2 & 8 & 2 \\ 1 & 7 & -1 \\ -2 & -2 & 1 \end{bmatrix}$

(b)  $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ -1 & 1 & 0 \\ 1 & 5 & 1 \end{bmatrix}$

2. Find the diagonalization,  $P^{-1}AP = D$ , of the matrix  $A$  and determine whether or not the columns of  $P$  are orthogonal.

(a)  $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$

(b)  $A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 3 & 1 \end{bmatrix}$

(c)  $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$